

Research Statement

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My research focuses on probabilistic techniques for analyzing stochastic processes and extending them to related models. In joint work with Brian Fralix, I developed a decomposition-based approach to computing the stationary distributions of discrete-time Markov chains using a certain class of measures, together with iterative and linear-system methods for computing those measures, with analogous continuous-time results under additional assumptions [6]. For upper-triangular Pólya Urn Processes, I extended the notion of a balanced urn from two colors to n colors, yielding explicit formulas for the Laplace transform of the number of balls at time t [8]. I then applied a related conditioning-based method to the Generalized Pólya Process and several of its multivariate extensions, deriving moment recursions and Laplace functionals by reducing the analysis to linear or Bernoulli ODEs instead of PDEs [7].

New Approach for Computing the Stationary Distribution

Let $\mathbf{X} = \{X_n : n \geq 0\}$ be a discrete-time Markov chain (DTMC) with transition matrix $\mathbf{P}$ and state space S . If $\mathbf{X}$ is irreducible and positive recurrent, then $\mathbf{X}$ has a stationary distribution $\boldsymbol{\pi}$ that can be found by solving the global balance equations $\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{P}$ with the normalization condition $\sum_{x \in S} \pi(x) = 1$. However, when the state space S is large or countably infinite, solving this system of equations directly is computationally infeasible, motivating iterative or other numerical methods to approximate the distribution. In joint work with Brian Fralix, I develop a decomposition-based approach to calculating the stationary distribution in terms of a certain class of measures. We further introduce an iterative scheme whose iterates increase component-wise to the associated measures [6]. I focus on the discrete-time case below, but analogous continuous-time results hold under a uniform boundedness assumption on sojourn rates.

Choose a finite, non-empty, proper subset $M \subset S$ and let $E = S \setminus M$. We regard M as a distinguished subset of states. For each $x \in M$, we construct a measure $\boldsymbol{\mu}_M^{(x)}$ that solves the global balance equations restricted to E . For $x \in M$ and $y \in S$, $\mu_M^{(x)}(y)$ is defined as the expected number of visits to y before the first positive return to M , starting from x . When M is a single state $\{x\}$, $\mu_{\{x\}}^{(x)}$ corresponds to the usual invariant measure associated with x . The random product technique [4] uses irreducibility to show that these measures satisfy the restricted balance equations. Moreover, under positive recurrence, these measures have finite mass.

We further establish that for irreducible and positive recurrent DTMCs, the collection $\{\boldsymbol{\mu}_M^{(x)} : x \in M\}$ forms a basis for the set of absolutely convergent solutions to the restricted balance equations on E . This results in the restriction of the stationary distribution $\boldsymbol{\pi}$ to E being a linear combination of these measures. In particular,

$$\pi(y) = \sum_{x \in M} \pi(x) \mu_M^{(x)}(y), \quad y \in E.$$

The coefficients $\{\pi(x) : x \in M\}$ can be found up to a common scalar multiple by solving the $|M| \times |M|$ linear system

$$\pi(x) = \sum_{y \in E} \sum_{z \in M} \pi(z) \mu_M^{(z)}(y) p(y, x) + \sum_{z \in M} \pi(z) p(z, x), \quad x \in M$$

where $p(y, x)$ is the (y, x) -entry of $\mathbf{P}$, and the scalar is fixed by the normalization condition

$$\sum_{x \in M} \pi(x) + \sum_{y \in E} \sum_{x \in M} \pi(x) \mu_M^{(x)}(y) = 1.$$

We also prove that each measure $\mu_M^{(x)}$ is the minimal non-negative solution to the restricted balance equations on E with prescribed values on M . The proof of this fact yields an iterative scheme that can be used to compute these measures. This scheme has component-wise non-decreasing iterates and converges even for periodic chains, whereas the standard power iteration need not converge for periodic chains. However, our theoretical results are much broader than this iterative scheme. These measures also satisfy the linear system

$$\mu_M^{(x)}[E](\mathbf{I} - \mathbf{P}[E, E]) = \mathbf{P}[\{x\}, E], \quad x \in M$$

where $\mu_M^{(x)}[A]$ represents the subvector of $\mu_M^{(x)}$ with entries indexed by A and $\mathbf{P}[A, B]$ represents the submatrix of $\mathbf{P}$ with rows indexed by A and columns indexed by B . When $\mathbf{X}$ is irreducible on a finite state space, the substochastic $\mathbf{P}[E, E]$ has a spectral radius less than 1. Hence $\mathbf{I} - \mathbf{P}[E, E]$ is nonsingular and the linear system for $\mu_M^{(x)}$ can be addressed using standard direct methods or appropriately chosen iterative solvers. This means our representation for π is not tied to our particular monotone iteration.

Upper-Triangular Pólya Urns

The Pólya urn Process is a continuous-time embedding of the discrete-time Pólya urn, obtained by associating with each object in the urn an exponential random clock with rate 1 independent of all the other objects' clocks. The clock ringing corresponds to the time when an object is drawn from the urn. This notion of embedding was introduced by Athreya and Karlin in [1].

Suppose we have n types of objects labeled 1 through n . The upper-triangular Pólya urn is a Pólya urn where when an object of type k is drawn, the object is placed back into the urn and more objects of type k through n are added to the urn. This can be represented by the $n \times n$ replacement matrix

$$\mathbf{A} = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} & \cdots & a_{1,n} \\ 0 & a_{2,2} & a_{2,3} & a_{2,4} & \cdots & a_{2,n} \\ 0 & 0 & a_{3,3} & a_{3,4} & \cdots & a_{3,n} \\ 0 & 0 & 0 & a_{4,4} & \cdots & a_{4,n} \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \ddots & a_{n,n} \end{pmatrix}.$$

For $i \leq j$, $a_{i,j} > 0$ denotes the amount of type- j material added when a type- i object is drawn. For classical urns, the entries $a_{i,j}$ are integer amounts and correspond to adding different colors of balls. More generally, we allow $a_{i,j}$ to be real-valued, in which case the objects in the urn correspond to containers of n different types of fluid. In the fluid interpretation, a container holding δ units of type- i fluid has an exponential clock with rate δ independent of every other container. When the container's clock rings, the container stays in the system, its clock is restarted, and for each $j = i, i+1, \dots, n$, an additional container of $a_{i,j}$ units of type- j fluid is added.

In [5], the authors address finding the joint moment generating function for the number of balls at time t in the case of 2 colors. Define $F_i(t)$ as the number of color i balls at time t in the classical urn model, or the amount of type- i fluid at time t in the fluid variation. Let $\mathbf{F}(t) = (F_1(t), F_2(t), \dots, F_n(t))$. In joint work with Brian Fralix, we derived a recursive expression for the joint Laplace transform of $\mathbf{F}(t)$ [8]. To establish this, we considered the Laplace transform for an urn that starts with only one type of fluid in it and conditioned on the time of the first arrival. This conditioning yields a linear or Bernoulli ODE, which can be solved recursively starting with type n . The resulting expression for the transform associated with type k fluid involves integrals of the transforms associated with types $k+1, k+2, \dots, n$. The branching property of the process allows us to combine the single-type transforms to obtain the transform for a general initial configuration. This approach avoids the PDE calculation used in the two-type analysis in [5] and extends naturally to the n -type setting.

I also extended the definition of a balanced urn from two types to n types, following the two-color notion introduced in [3]. The authors of [3] define the notion of a balanced urn for two colors: the same number of balls are always added to the urn each time a ball is drawn, regardless of the color. In the fluid interpretation, this means the same total volume of fluid is added. In terms of the replacement matrix, this means $a_{2,2} = a_{1,1} + a_{1,2}$ and $a_{1,1} < a_{2,2}$. I extended this notion of balance for two colors to the general n types of fluid case in the following way: for each $k = 1, 2, \dots, n-1$,

- (a) $a_{k,k} < a_{k+1,k+1}$;
- (b) $a_{k+1,k+1} = a_{k,k} + a_{k,k+1}$;
- (c) for each $i = 1, 2, \dots, k$, $a_{i,k+1} = a_{k,k+1}$.

These conditions imply that every row of the replacement matrix has the same sum, namely $a_{n,n}$, preserving the original interpretation of balance. They also allow the integral terms in the recursive Laplace-transform representation to be evaluated explicitly, yielding an explicit formula in terms of exponential functions.

Generalized Pólya Processes

The conditioning approach used for the upper-triangular Pólya urn also applies to the Generalized Pólya Process. The Generalized Pólya Process (GPP) is a self-excited point process that behaves in the following way. Begin with a non-homogeneous Poisson process (NHPP) with rate function $\beta\kappa(t)$, where $\beta, \gamma > 0$ and $\kappa : [0, \infty) \rightarrow [0, \infty)$ is a continuous function. Each arrival independently initiates an offspring NHPP with rate $\gamma\kappa(t)$ for t greater than the arrival time; every subsequent arrival initiates another such process. Let $\{N(t) : t \geq 0\}$ denote the GPP with characteristic triple (β, γ, κ) .

In joint work with Brian Fralix, we use the same basic conditioning approach to derive the Laplace functional of the GPP [7]. We first analyze a truncated version of the Laplace functional on the interval $(s, t]$ for a non-negative continuous function f with compact support. We condition on the time of the first arrival of the GPP. Conditional on the first-arrival time, the descendants generated by that arrival form an independent GPP with triple (γ, γ, κ) , while the remaining contribution is represented by the (β, γ, κ) process. This conditioning leads to a linear ODE whose solution involves the truncated Laplace functional for the (γ, γ, κ) process; conditioning once more yields a Bernoulli ODE for that functional.

Solving the Bernoulli ODE and combining with the result for the (β, γ, κ) process, we set $s = 0$ and let $t \rightarrow \infty$ to obtain the Laplace functional. Because f has compact support, the truncated functional stabilizes once t exceeds the support of f . We obtain that the Laplace functional is

$$\mathbb{E}[e^{-\int_{[0,\infty)} f(s)N(ds)}] = \left(\frac{(1/\gamma)}{(1/\gamma) + \int_0^\infty (1 - e^{-f(u)})\kappa(u)e^{\gamma \int_0^u \kappa(x)dx} du} \right)^{\beta/\gamma}.$$

Here, $N(ds)$ represents the counting measure associated with the GPP $\{N(t) : t \geq 0\}$. Since the Laplace functional uniquely characterizes a point process, this formula provides a complete distributional characterization of the GPP.

We apply the method more broadly to obtain moment recursions and the Laplace functional for a two-function generalization of the GPP studied in [9], as well as Laplace functionals for the multivariate extensions of the GPP in [12] and [11]. We also develop an upper-triangular multivariate GPP that generalizes the combined imperfect repair process in [2] and derive its Laplace functional. For this model, the offspring-arrival mechanism is the same as in the univariate GPP, but the offspring types are assigned according to time-dependent probabilities. For every $k = 1, 2, \dots, n$ and $\ell = k, k+1, \dots, n$, let $p_{k,\ell} : [0, \infty) \rightarrow [0, 1]$ be functions satisfying

$$\sum_{\ell=k}^n p_{k,\ell}(s) = 1 \text{ for all } s \geq 0.$$

Conditional on being born at time s from a type- k point, an offspring is assigned type $\ell \in \{k, k+1, \dots, n\}$ with probability $p_{k,\ell}(s)$. Conditional on their birth times and parent types, offspring types are assigned independently according to these probabilities. Thus, offspring types can remain their parent's type or move to a higher type (never lower), and the type-transition probabilities may vary with time. These results show that the conditioning method extends beyond the univariate GPP to models with more general parameter functions and multivariate branching structure.

Future Work

My principal future directions are computational methods for Markov chains and extensions and applications of the GPP. I also plan to pursue a smaller project on other Pólya urn models. For Markov chains, I will determine when the Markov chain framework discussed above can be turned into a computationally competitive algorithm for computing the stationary distribution. I will first analyze our new iterative scheme to establish or estimate the rate of convergence and compare its performance with the standard power iteration and other solvers. Then, I will develop an algorithm for computing the stationary distribution based on the measure representation. Even when our iterative scheme is not the fastest method, the measure representation provides an alternative formulation to which established solvers can be applied. When $|M| = 1$, the algorithm reduces to computing one invariant measure and normalizing it. When $|M| > 1$, one must compute $|M|$ measures and solve an additional $|M| \times |M|$ system. I will determine for which classes of chains the parallel computation of the $|M|$ vectors, along with favorable sparsity or block structure in $\mathbf{P}$, can offset the extra work when $|M| > 1$.

For the GPP, I will investigate how far the continuity assumption on κ can be weakened while preserving the Laplace functional formula found above. I will investigate extensions from continuous rate functions to broader classes of nonnegative locally integrable functions, under

conditions ensuring that the relevant integrals remain finite and the Laplace-functional formula remains valid. I also plan to develop numerical methods for evaluating the moment recursions for the multivariate GPP models, with the goal of using the resulting moments to compute reliability measures for models such as the combined imperfect repair process studied in [2], as well as for related models such as our upper-triangular multivariate GPP.

I also plan to extend the conditioning-based approach to other Pólya urn models. I have already used this approach to derive moment recursions and Laplace transforms for several other urn models. However, for the Apollonian urn studied in [10], the corresponding Laplace-transform calculation does not reduce to a comparably simple ODE. I plan to investigate whether techniques from branching-process theory can exploit the branching structure of this urn to produce a tractable transform representation, with the broader goal of identifying which structural features of an urn permit explicit Laplace-transform formulas.

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